

# Arbaz Shareef Mohammed

Senior Python / Full Stack Developer | AI/ML & Cloud Specialist

✉ mohammedshareefarbaz@gmail.com | ☎ 484-620-0950

🔗 LinkedIn: <http://linkedin.com/in/arbaz-shareef-mohammed-550967281>

---

## PROFESSIONAL SUMMARY

Senior Python Developer with **10+ years of experience** designing and delivering scalable, high-performance web applications, cloud-native platforms, and end-to-end data pipelines across finance, healthcare, and e-commerce domains. Strong expertise in Python, microservices, cloud infrastructure, AI/ML integration, and DevOps automation. Proven ability to lead complex projects, optimize performance, and deliver business-critical solutions.

---

## CORE SKILLS

**Programming:** Python, JavaScript, Shell Scripting

**Frameworks:** Django, Flask, FastAPI, Django REST Framework

**Frontend:** React, Next.js, Angular, AngularJS, Redux, HTML5, CSS3, Bootstrap

**Cloud:** AWS, Azure, GCP

**Data Analytics & BI:** Power BI, Power BI Service, DAX, Dataflows, Row-Level Security (RLS), Star Schema Modeling, Incremental Refresh, Executive & KPI Dashboards

**DevOps:** Docker, Kubernetes, Jenkins, GitLab CI/CD, Terraform, CloudFormation, Ansible

**Databases:** PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, DynamoDB, Redshift, Cassandra

**AI / ML:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, GPT-3, BERT, RoBERTa

**Data & Streaming:** Apache Spark, Kafka, Airflow, Hadoop

**Security:** OAuth2, JWT, RBAC, OpenID Connect, PCI-DSS

**Tools:** Git, Postman, Swagger, Power BI, Tableau, JIRA, Confluence, UML

---

## DATA ANALYTICS & POWER BI EXPERIENCE

- Designed and delivered **Power BI dashboards** for executive leadership, compliance teams, and operations stakeholders, enabling data-driven decision-making across finance and government programs.
- Built enterprise-grade **Power BI data models** using star schema design, optimized relationships, and **incremental refresh** for datasets exceeding **10M+ records**.
- Authored advanced **DAX measures** for time intelligence, rolling averages, SLA tracking, utilization forecasting, anomaly detection, and KPI benchmarking.
- Implemented **Row-Level Security (RLS)** to enforce role-based access control across analyst, manager, and executive views.
- Developed reusable **Power BI Dataflows** to standardize transformations and reduce dataset refresh time by **30-40%**.

- Integrated Power BI with **Azure SQL, BigQuery, PostgreSQL, and Redshift**, supporting near real-time reporting.
  - Reduced manual reporting effort by **40%** and improved dashboard performance by **35%** through query optimization and model tuning.
- 

## PROFESSIONAL EXPERIENCE

### Senior Python Developer

**PennyMac – Westlake Village, CA | Aug 2023 – Present**

- Designed and delivered cloud-native, microservices-based applications using **Python, Flask, and Django REST Framework** to process and analyze large volumes of unstructured financial and compliance documents.
- Led the transition of document monitoring and analytics services to a **cloud-native architecture**, improving scalability, reliability, and deployment velocity.
- Built real-time ingestion pipelines using **Apache Kafka**, enabling near real-time processing of regulatory filings, research documents, and external data feeds.
- Integrated **Azure OpenAI (GPT-3)** and **Hugging Face Transformers** to generate semantic embeddings, summaries, and intelligent document search capabilities beyond keyword-based retrieval.
- Designed and implemented secure **RESTful APIs** with **OAuth2, JWT, and RBAC**, supporting role-based access for analysts, compliance officers, and administrators.
- **Designed, developed, and maintained Power BI dashboards** for executive leadership, compliance, and risk teams to monitor document ingestion volumes, SLA adherence, exception trends, and regulatory KPIs.
- Built enterprise-scale **Power BI data models** using star schema design, optimized relationships, and **incremental refresh** to support datasets with **millions of financial records**.
- Authored advanced **DAX measures** for time intelligence, rolling averages, month-over-month comparisons, anomaly detection, and compliance trend analysis.
- Implemented **Row-Level Security (RLS)** in Power BI to enforce strict role-based access controls aligned with compliance and audit requirements.
- Integrated Power BI with **Azure SQL, BigQuery, and REST APIs**, enabling near real-time reporting and automated executive dashboards.
- Reduced manual compliance reporting effort by **40%** and improved dashboard performance by **35%** through data model and query optimization.
- Containerized services using **Docker** and deployed to **Azure Kubernetes Service (AKS)** with Helm, enabling zero-downtime deployments and auto-scaling.
- Implemented robust **CI/CD pipelines** using GitLab CI/CD, automating testing, image builds, and deployments across environments.
- Collaborated closely with product, compliance, data, and leadership teams to translate regulatory requirements into scalable technical and analytical solutions.

**Environment:** Python, Flask, Django REST Framework, Kafka, Azure OpenAI, Hugging Face, Power BI, DAX, Azure SQL, BigQuery, Docker, AKS, Helm, GitLab CI/CD

---

## Senior Python Developer

**State of Virginia – Richmond, VA** | *Mar 2021 – Jul 2023*

- Designed and implemented scalable backend services using **Python, Django REST Framework**, supporting ingestion and processing of high-volume real-time telemetry data from machine sensors across multiple regions.
- Built real-time telemetry ingestion pipelines using **Spark Streaming and PySpark**, processing millions of events daily with low-latency guarantees.
- Developed full-stack fleet monitoring dashboards using **Angular and Django**, enabling operations and leadership teams to track machine health, utilization, downtime, and alerts in near real time.
- Designed and delivered **Power BI dashboards** for executive leadership, providing visibility into fleet utilization trends, sensor health KPIs, SLA compliance, and operational efficiency.
- Built optimized **Power BI data models** using star schema design and incremental refresh, improving dashboard load times and scalability for large telemetry datasets.
- Authored advanced **DAX calculations** for rolling averages, utilization forecasting, time-based comparisons, and performance benchmarking.
- Developed and trained **machine learning models** using Scikit-learn to predict equipment failures and support proactive maintenance strategies.
- Automated complex **ETL workflows using Apache Airflow**, implementing retries, alerting, and SLA monitoring to ensure data reliability.
- Collaborated with cloud and DevOps teams to deploy services on **AWS and Azure Kubernetes Service (AKS)**, ensuring high availability and fault tolerance.
- Actively participated in Agile ceremonies, sprint planning, backlog grooming, and production support activities.

**Environment:** Python, Django REST, Angular, Spark, PySpark, Airflow, AWS, AKS, Power BI, DAX, Docker

---

## Senior Software Engineer

**UHG – Optum, MN** | *Jul 2019 – Feb 2021*

- Designed, developed, and maintained cloud-native healthcare applications using **Python and Django REST Framework**, supporting patient intake, scheduling, and care coordination workflows.
- Led migration of legacy on-premise applications to **AWS microservices architecture** using ECS, Lambda, and Fargate, improving system reliability and scalability.
- Built secure and compliant RESTful APIs handling sensitive healthcare data, adhering to **HIPAA compliance** standards.
- Automated infrastructure provisioning and deployments using **Terraform and Jenkins**, enabling repeatable and auditable releases across environments.
- Implemented asynchronous processing using **Celery and RabbitMQ** to support background jobs, notifications, and batch workflows.

- Integrated monitoring and observability using **New Relic, Datadog, and Splunk**, improving incident detection and mean-time-to-resolution.
- Optimized application performance using Redis caching and database query tuning.
- Collaborated closely with product managers, clinicians, and QA teams to deliver high-quality features on tight timelines.

**Environment:** Python, Django REST, React, AWS, Terraform, Jenkins, Redis, RabbitMQ

---

## Software Engineer

**UBS – Stamford, CT** | Oct 2017 – Jun 2019

- Contributed to the redesign of a monolithic fraud detection platform into **microservices-based architecture** using Flask and Django.
- Developed backend services to ingest, normalize, and analyze fraud signals from multiple external and internal data sources.
- Built interactive dashboards using **Angular and React**, enabling analysts to investigate alerts, risk scores, and fraud trends.
- Implemented real-time data streaming pipelines using **Apache Kafka** and scheduled workflows using **Apache Airflow**.
- Designed and optimized PostgreSQL and Redshift schemas to support high-performance analytical queries.
- Automated deployments using Docker and Jenkins, improving release frequency and stability.
- Worked closely with cybersecurity analysts and DevOps teams to enforce secure coding and access control practices.

**Environment:** Python, Flask, Django, Angular, React, Kafka, Airflow, AWS

---

## Python Full Stack Developer

**Barclays – Mumbai, India** | Sep 2014 – Aug 2017

- Developed enterprise-grade internal web platforms using **Python, Django, and AngularJS**, supporting operations, supply chain, and inventory management teams.
- Implemented scalable RESTful APIs and backend services using Django and Flask, handling high-volume transactional workflows.
- Designed and maintained **ETL pipelines** to ingest and transform large XML and CSV datasets from third-party vendors and internal systems.
- Processed structured data using **Apache Spark**, improving batch processing efficiency and reliability.
- Designed optimized **MongoDB and MySQL schemas**, implementing indexing and sharding strategies for performance.
- Containerized applications using **Docker** and automated CI/CD pipelines using **Jenkins**.
- Deployed services on **AWS EC2, S3, and Lambda**, supporting both server-based and serverless workloads.

**Environment:** Python, Django, AngularJS, MongoDB, MySQL, Spark, AWS\*\*.

**Environment:** Python, Django, AngularJS, MongoDB, MySQL, Spark, AWS

---

## **EDUCATION**

**Bachelor of Technology (B.Tech)**

**Electronics & Communication Engineering**